COMP2300 Set 4 Practice Exam Solutions

Part A: Multiple Choice

A1. B

A2. C

A3. C

A4. B

A5. B

A6. A

A7. A

A8. B

A9. B

A10. C

Part B: Computational Problems

B1. Endianness and PC update

(a) Little-endian word at 0x100: 0x78563412.

(b) Big-endian word at 0x100: 0x12345678.

(c) Next sequential PC = 0x104.

B2. Base + offset

(a) Address = 0x08, so $R3 = 0x11223344$.

(b) Store to 0x0C, so memory[0x0C] = 0xCAFEBABE.

B3. Flags and conditional execution

(a) $17 - 23 = -6$ so $N = 1$, $Z = 0$, $C = 0$ (borrow), $V = 0$.

(b) `SUBEQ` does not execute (Z=0). `ORRMI` executes (N=1). `BGE` does not execute (N $\neq$ V).

B4. Branch target address

PC+8 = 0x80A8. imm24 = 0x000003 so (imm24 $\ll$ 2) = 0x00000C.
BTA = 0x80A8 + 0x00000C = 0x80B4.

B5. Immediate extension

(a) ExtImm = 0x000000AB.

(b) ExtImm = 0x00000345.

(c) ExtImm = 0x0003C3C0.

B6. Performance

(a) Time = $N \times CPI/f = 1.2 \times 10^9 \times 1.8/2.4 \times 10^9 = 0.9$ s.

(b) 2 ns clock $\Rightarrow f = 500$ MHz. Time $= 5 \times 10^8/5 \times 10^8 = 1$ s.

Part C: Past Exam Questions

C1. RISC vs CISC

RISC uses a small set of simple, typically fixed-length instructions and a load/store model; CISC uses more complex instructions and richer addressing modes. ARM is a RISC ISA. QuAC is also a RISC-style teaching ISA.

C2. Branch prediction

B (Microarchitecture).

C3. Execution time

$T = 2 \times 10^9 \times 4/2 \times 10^9 = 4$ seconds.

C4. Address calculation

C. `LSL R1, R0, #4` (multiply index by 16).